

1 Question (25 points)

Describe a single tape TM that recognizes the language $L = \{0^n 1^{n^2} 2^{n^2} | n \geq 1\}$. Do not design the TM. Describe the steps it needs to take.

2 Question (25 points)

Describe a two tape TM that recognizes the language $L = \{0^n 1^{n^3} | n \geq 1\}$. Describe the steps of TM. You must use both tapes in your solution.